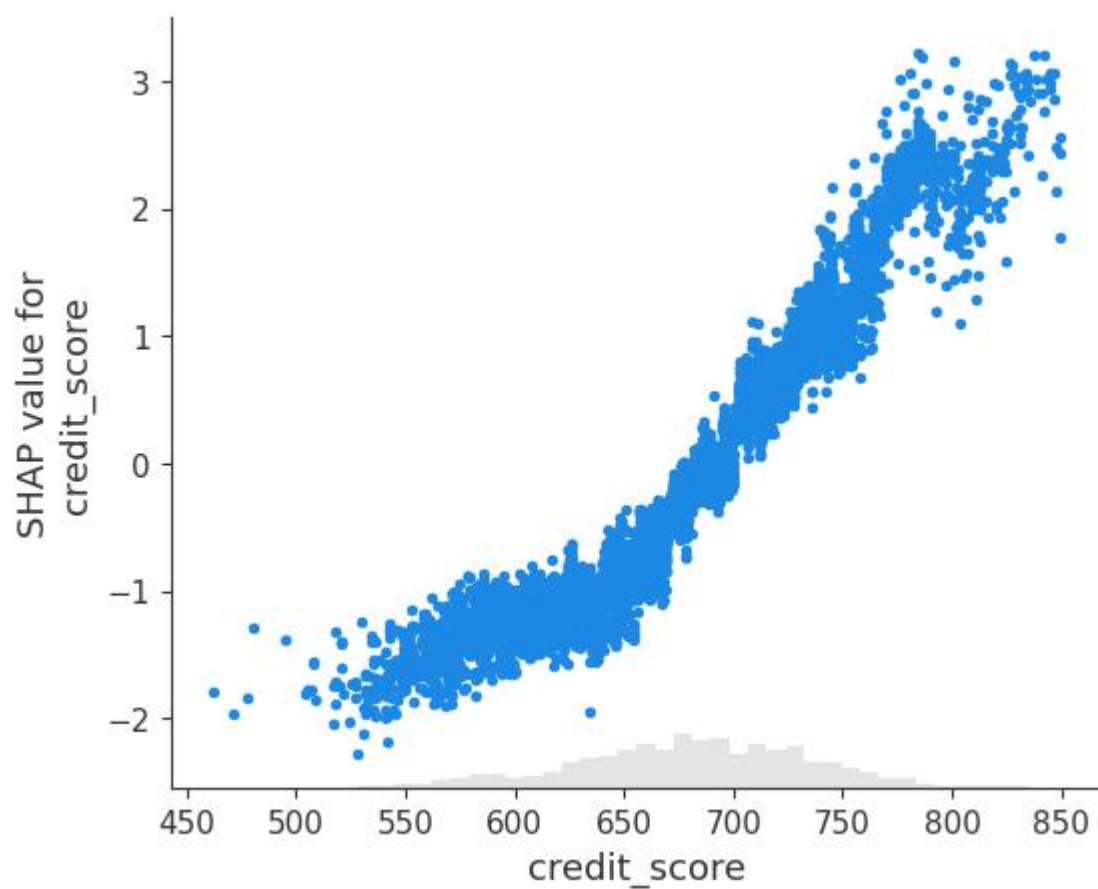
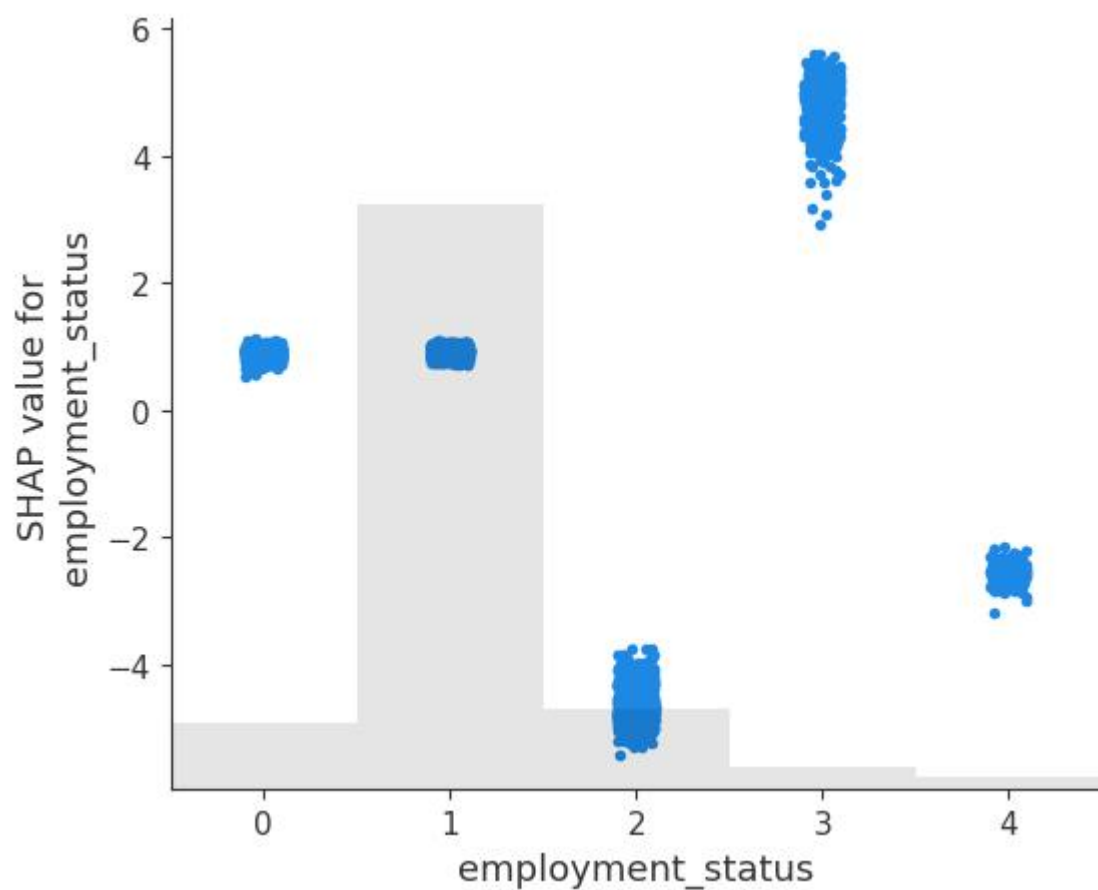
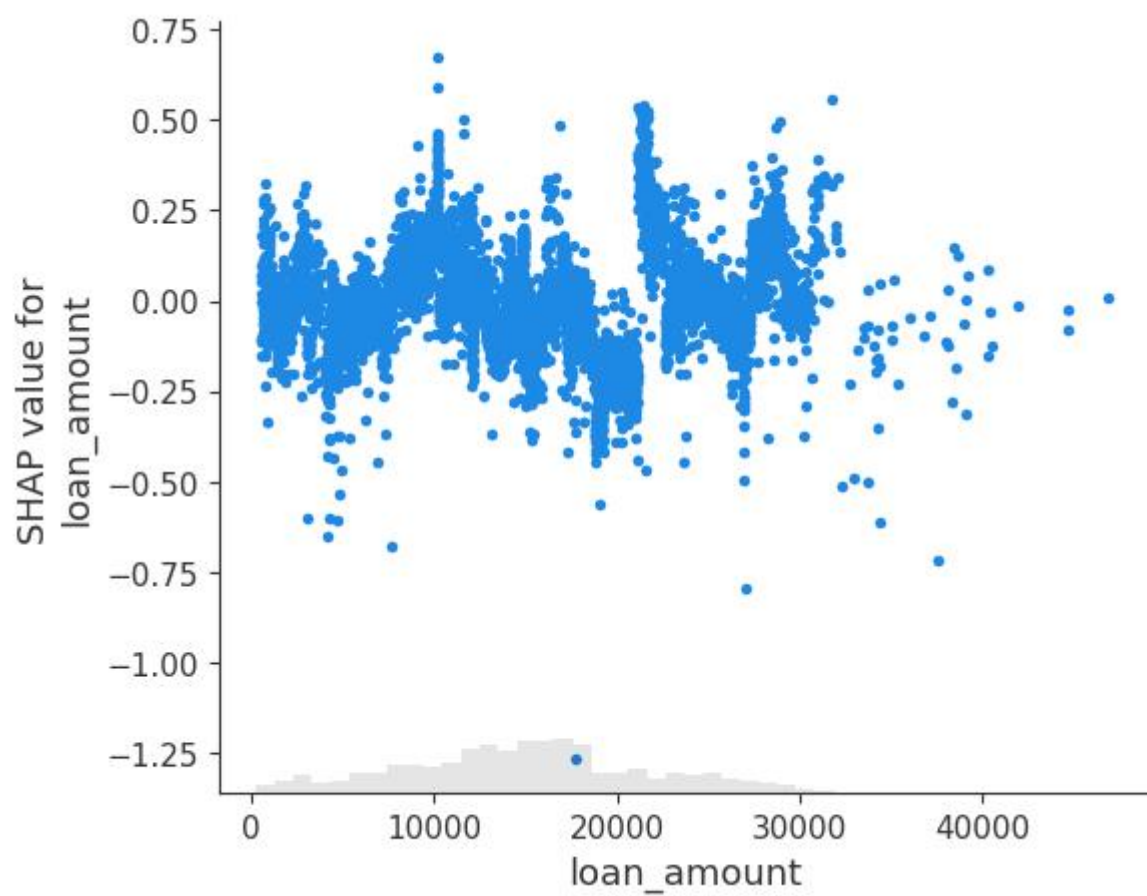
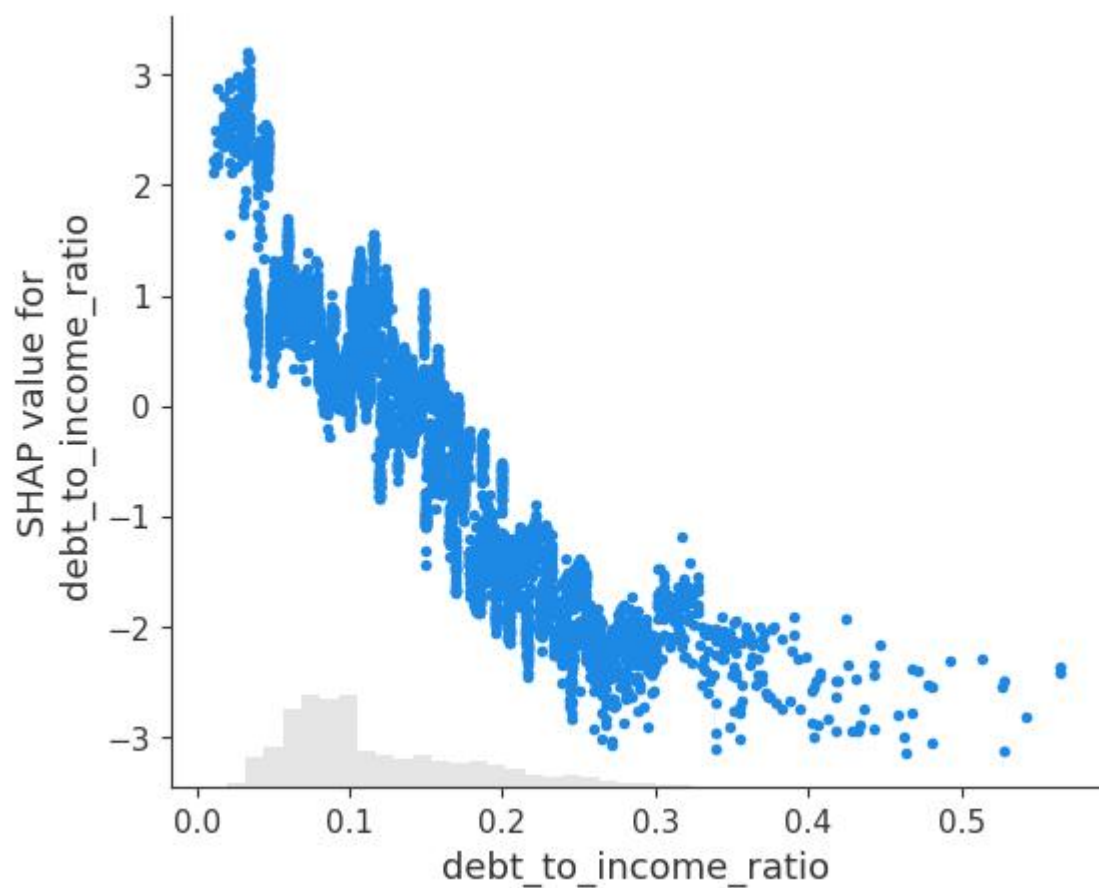
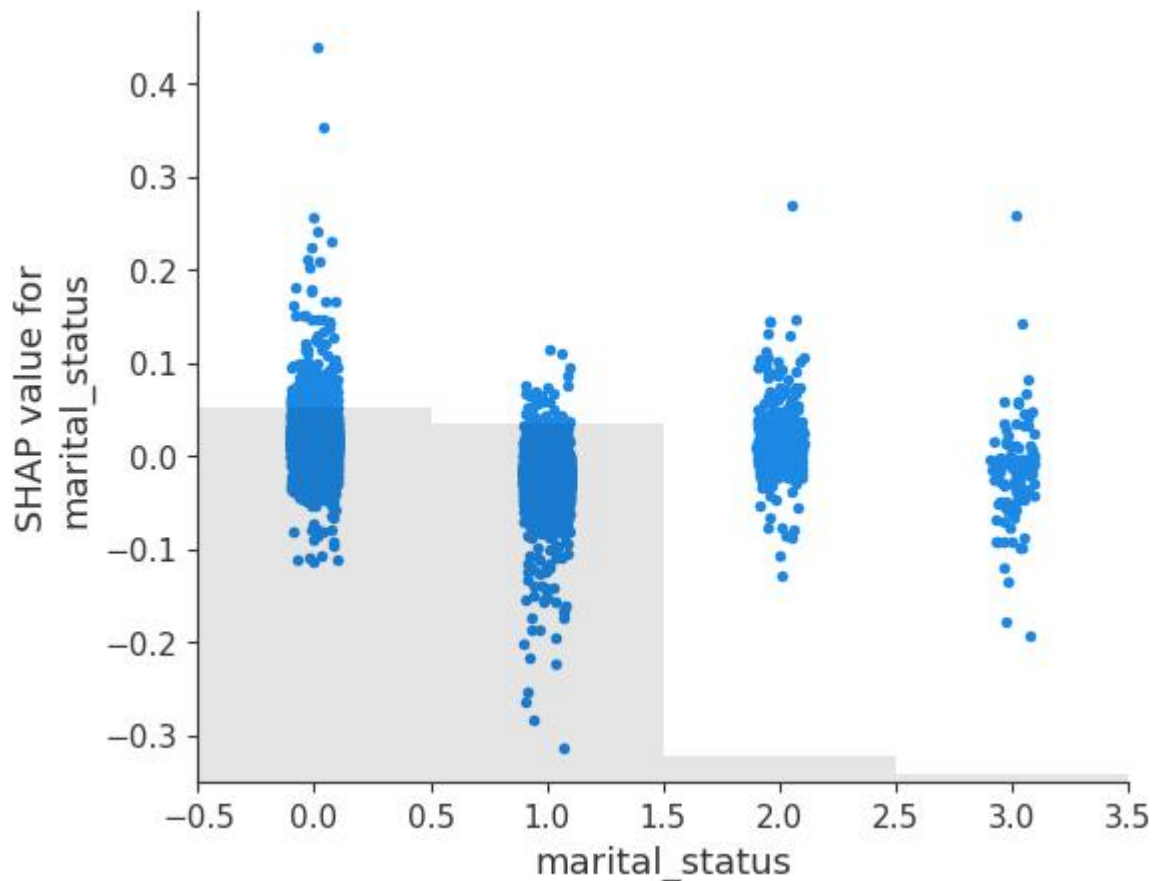








Encoding for categorical features:

Feature	Encoding	Category
employment_status	0	Self-Employment
	1	Employed
	2	Unemployed
	3	Retired
	4	Student
marital_status	0	Single
	1	Married
	2	Divorced
	3	Widowed

Gemini Analysis

Prompt:

What I have here is a scatter plot of SHAP values from a ensemble model. Some features is written (as is) as floats where some are encoded categoricals. Categorical features and their encoding are available on the table at the last part of this pdf. Interpret the model based on these plot.

Result:

Based on the SHAP scatter plots provided, here is an interpretation of how the ensemble model utilizes these features to make predictions. SHAP values indicate how much a specific feature value contributes to the model's output compared to the average prediction; positive values increase the prediction, while negative values decrease it.

1. Categorical Feature Analysis

The model shows distinct preferences based on the encoding provided in the table

- **Employment Status:** This is a major driver for the model

- **Retired (3):** Has the strongest positive impact, with SHAP values between +4 and +6.
- **Employed (1) & Self-Employed (0):** Both show a moderate positive impact, clustering around \$+1\$.
- **Unemployed (2):** Has a significant negative impact, dropping SHAP values to approximately -4 to -5.
- **Student (4):** Shows a moderate negative impact, around -2.5.
- **Marital Status:** This feature has a **very low impact** on the model. All categories (Single, Married, Divorced, Widowed) result in SHAP values very close to 0, ranging roughly between -0.2 and +0.2.

2. Numerical Feature Trends

The continuous features show clear non-linear relationships with the model's output.

- **Credit Score:** There is a strong **positive correlation** between credit score and SHAP values
 - The "neutral" point (SHAP = 0) occurs around a score of **700**
 - Scores above 750 significantly boost the prediction (up to +3 SHAP)
 - Scores below 600 penalize the prediction (down to -2 SHAP).
- **Debt-to-Income (DTI) Ratio:** There is a clear **negative correlation**.
 - Low DTI ratios (below 0.15) provide a strong positive contribution (up to +3 SHAP)
 - As the ratio increases beyond 0.2, the impact becomes increasingly negative, bottoming out around -3 for ratios above 0.35
- **Loan Amount:** This feature has a **low and noisy impact**.
 - Most SHAP values fluctuate near 0 (between -0.5 and +0.5) regardless of the loan size.
 - The model does not seem to heavily favor or penalize specific loan amounts within the 0 to 40,000 range.

3. Summary of Model Behavior

Feature	Impact Level	Direction
Employment Status	High	Favors Retired/Employed; Penalizes Unemployed.
Credit Score	High	Strong positive correlation; 700 is the pivot point.
DTI Ratio	High	Strong negative correlation; favors low ratios.
Loan Amount	Low	Minimal impact on the final decision.
Marital Status	Negligible	Almost no influence on the model.

The model appears to prioritize **financial stability** (Credit Score, DTI) and **income reliability** (Employment Status) over demographic factors (Marital Status) or the specific size of the request (Loan Amount).